# Anton Grishin — Backend Developer


## Personal Information

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## About Me

I am a student and an aspiring developer. I have been programming for 4 years, during which I contributed to 44 repositories, made 921 commits, and wrote 176,729 lines of code. Former professional volleyball player. I am also a Python instructor, having taught over 30 students. Co-founder and developer at Ballkit.

## Skills

**Python:** `FastAPI`, `pydantic`, `Flask`, `SQLAlchemy`, `faststream`, `alembic`, `pytest`.

**Go:** `http`, `grpc`, `protobuf`, `tgbotapi`, `reflect`, `gofsm`.

**Databases:** `postgres`, `sqlite`, `redis`, `Yandex Object Storage`.

**Message brokers:** `RabbitMQ`, `Mosquitto`.

**Other technologies:** `SQL`, `Java`, `JavaScript`, `bash`.

**Dev tools:** `Docker`, `Podman`, `Make`, `CI/CD`, `Linux`, `Git`

## Education

Central University — Mathematics and Computer Science, 2028

## Certifications

**Yandex Lyceum**. Admitted externally to the 2nd year of the Yandex Academy Lyceum in "Industrial Development."Solved 236 problems, implemented and defended 3 projects, scored 100/100 on the final. Graduated with honors.

*September 2022 — April 2023*

**Python Course**. Completed the Stepik course on `Python`, solving 71 exercises and finishing with distinction.

*September 2021 — April 2022*

## Achievements

Finalist of the **Russian Championship** in competitive programming (among 5000 participants). Developed a microservice architecture of 9 services for a web application aggregating sports events across Russia. Also created an algorithm for processing and validating annual state reports on sports events. Tech stack: `Kafka`, `React`, `RabbitMQ (FastStream)`, `FastAPI`, `OAuth`.

*December 2024*

**Winner** of the "Science for Life"scientific-practical conference among 117 projects with a smart home project for private and public educational institutions. Tech stack: `Redis`, `Zigbee2MQTT`, `websockets`, `Go`, `Python`, `Flask`, `React`.

*April 2024*

Participant in the **Nuclear IT Hack** hackathon, where our team worked for 48 hours on a Rosatom case: a service for detecting emotional tone in online meetings.

*April 2024*

**3rd Prize** in the MPGU Olympiad in Informatics: "Applied Informatics".

*June 2023*

## Interests

**Formal Verification:** Currently taking the Software Foundations course. Completed the first volume and proved 250 theorems in `Coq`.

**Linux:** Using Arch with the Hyprland compositor. Published my .dotfiles. Wrote my Neovim config from scratch and a color theme. Authored 598 lines of custom scripts.

## Startup Experience

**Patch Loyalty** Loyalty program for SMBs *July 2024 — August 2024*

Developed a Single Message Bot in Telegram using `Go` for the boxed loyalty solution `ballkit.ru`.

**Technologies:** `Go`, `tgbotapi`, `grpc`, `gofsm`.

## Personal Projects

**SmartCab** School smart home system *September 2023 — May 2024*

Designed admin panel and tablet interface prototypes and implemented them in `React`. Developed a bot for system administration and implemented interaction with five smart devices over Zigbee via Mosquitto broker and three classroom scenarios. Technologies: `Python`, `Go`, `JavaScript`, `bash`, `Flask`, `sqlite`, `redis`, `postgres`, `JsSip`.

**Starship** Text-based action game in Telegram *March 2023 — April 2023*

Developed real-time duel mechanics for players in a Telegram bot and a flexible message rendering engine using jinja2 templates for complex narrative texts. Technologies: `Python`, `Flask`, `alembic`, `python-telegram-bot`, `SQLAlchemy`, `jinja2`.

**Corpse** Game engine *November 2022 — January 2023*

Collaborated on a modular ECS-based game engine and built a survival top-down shooter about a zombie apocalypse. Technologies: `Python`, `Pygame`, `esper`, `pytmx`, `aseprite`, `Tiled`.

**SchoolRing** School bell scheduling `release v0.0.2` *October 2022 — November 2022*

Developed a desktop application in `PyQt6`. Configured `pre-commit` hooks to auto-convert .ui files to .py and CI actions for cross-platform releases. Recently migrated the project to `uv` for dependency management. Technologies: `Python`, `PyQt6`, `threading`, `schedule`.

## Contract Development

**Telegram Bots** *Publication of all sources approved* *Summer 2024*

- Developed a bot for administering and selling WireGuard keys via subscription. Implemented inline analytics reports directly in the bot. Technologies: `Python`, `sqlite`, `python-telegram-bot`.
- Bot for aggregating relevant messages from chats, channels, and groups by keywords. Technologies: `Python`, `theleton`, `redis`

**Other Projects** *Publication of all sources approved* *Summer 2024*

- Parser for the Portuguese embassy website with authentication `auth.py`, captcha solving `captcha.py`, and branching form traversal `snif.py`. Created a Docker image for containerized parsing using Selenium on Chromium. Configured `pre-commit` hooks for Ruff linter and Pyright static analysis. Technologies: `Python`, `selenium`, `psycopg`, `SQLAlchemy`, `FastAPI`.
- CLI tool for secure operations on Outline servers across thousands of clients. Technologies: `Python`.

## Lab Works

**BookSwap** Book exchange service *May 2024*

Designed the database schema definition. Implemented the backend with core business logic in `Spring` and the authentication service in `Go`, plus a frontend in `React`.

**Technologies:** `Java`, `SpringBoot`, `Lombok`, `JavaScript`, `React`, `Go`, `crypto`, `pgx`.

→ more projects on GitHub